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Group - 29



ROAD SURFACE DAMAGE DETECTION

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INTRODUCTION OF ROAD INFRASTRUCTURE

Road networks are a fundamental part of the transportation system associated with features crucial for moving people and goods, as well as supporting economic activities. Roads connect communities, and smooth transportation flow.

Road problems in our country refer to various challenges such as: road accidents, bad road conditions, potholed roads, lack of road safety, traffic congestion, surface wear and erosion, rutting, poor drainages, etc...

ROLE OF IMAGE & VIDEO PROCESSING

Current advance in camera technology and computer vision have made it possible to develop **Road Surface damage detection** Project alternative to traditional quality assessment methods.

We are presently working on our road surface damage detection project. Members of our team are also actively Discuss and Planning about how we would implement our project. The following are the main implementation methods suggested for the project.

METHODS

Phase 1 – Image Acquisition Model

We will make use of our personal mobile phones use to capture video. In our case, we will ensure that there is little or no shadows. Water-filled potholes will also be avoided in order to ensure that there is a clear view of the damage. Important variables like time of filming, positioning, lighting, phone capabilities, and weather will also be taken into consideration. In addition, filming will take place both in stationary and walking positions.

Phase 2 – Image Enhancement Techniques

Noise, blurriness, changes in light intensity, and other distortions in the acquired video frames are identified and analysed. Individual frames of the video are extracted since damage detection in video frames is easier and accurate. Irrelevant or blurred video frames are then discarded with the purpose of enhancing data quality. Once all processes in this step are done, the system moves to the evaluation step.

Phase 3 – Evaluation and Segmentation

The system identifies the specific regions of the damaged road surfaces in the captured videos. During the running of the video, various regions of the damaged road surfaces are identified and marked in the videos. The base video and the marked videos are produced separately. The system then applies the proposed algorithm to classify the identified regions to determine the kind of damaged regions of the road surfaces in the video.

OBJECTIVE OF THE PROJECT

The primary aim or objective of this project is to minimize road related accidents and enhance human safety by identifying dangerous road conditions.

This our system captures video data of roads and processes identify and marked damaged areas such as: Cracks, Potholes, Edge Cracking, Pedestrian Crossing Marking Removal

When road damage is detected, the system can produce report and we can notify the relevant government authorities so that timely corrective action can be taken

In addition to that, this project offers educational benefits by demonstrating how image and video processing concepts can be applied to real world problems.